

Localization in 3-D Using Beacons of Low Frequency Magnetic Field

Arie Sheinker, *Member, IEEE*, Boris Ginzburg, Nizan Salomonski, *Member, IEEE*, Lev Frumkis, and Ben-Zion Kaplan, *Life Senior Member, IEEE*

Abstract—Traditional localization techniques, such as radar and GPS, rely on RF waves, which require line-of-sight for effective operation. In contrast, applications based on low frequency magnetic fields benefit from high penetration ability to crown canopy, soil, and many other types of media. In previous work we introduced a method for 2-D localization using beacons of low frequency magnetic field. Here, we propose to expand the method for 3-D localization. A mathematical analysis of the beacons' magnetic fields results in closed-form formulas which enable simple localization calculations. The method has been tested using numerous computer simulations showing accurate localization results in noisy environment and in various beacon configurations. A field prototype of the system has been developed and tested in field conditions, validating simulation results. The obtained experimental results show that the mean localization error is smaller than 0.25 m, and the maximal localization error is less than 0.77 m. The simple implementation, together with the fair localization accuracy, make the proposed method attractive for many applications requiring field penetration ability, such as indoor robot navigation, underground cavity mapping, and many more.

Index Terms—Beacons, localization, low frequency magnetic fields, magnetometer, search-coil sensor.

I. INTRODUCTION

LOCALISATION with the help of low frequency magnetic fields turns to be attractive in case where high penetration ability is required. It does not suffer from multipaths effects, and is unaffected by moisture and bad weather conditions [1]. Besides the aforementioned advantages over higher frequencies, using the ULF band (0.3–3 kHz) has also advantages over lower frequencies [2], which suffer from much higher magnetic background and interferences caused by power lines and electrical equipment [3]. Using low frequency magnetic fields has also several restrictions. The main restriction is a rather short operational range [4] relative to other localization techniques such as radio location. In addition, metals and interfering fields may distort localization results

[5]–[7]. Nevertheless, many applications such as localization [8], underwater communication [9], geophysical exploration [6], and identification of objects tagged by RFID [10], take advantage of the penetration ability of low frequency magnetic fields. Focusing on localization, a partial list of major applications, which employ low frequency magnetic fields, includes: underground geo-location and cavity mapping [11], [12]; zoological applications including revealing the hidden lives of underground animals such as badgers [13]; motion tracking, virtual and augmented reality facilities [14], [15]; medical applications including pill tracing, endoscopy, dosage transition monitoring [16]; localization of trapped miners and other human rescue applications [17]; indoor autonomous robot navigation and other navigation applications [18]; and location of buried objects in a conductive soil and UXO disposal applications [19]. Each of the above-mentioned methods features its specific drawbacks such as: short range, low accuracy, high computational burden, and necessity in multipoint measurements.

In previous work, we have introduced a method for 2-D localization based on a dual-axis search coil magnetometer [20], using beacons of low frequency magnetic field [21]. Here, we propose to expand the method for 3-D localization.

Relative to aforementioned techniques, the proposed method combines together the following important features:

- 1) Localization is done using only a pair or a triple of beacons, which may be concentrated in a small area relative to the search area.
- 2) The proposed localization method relies only on beacons with horizontal magnetic moment, which produce a magnetic field that does not feature additional decay over conductive soils [22]–[24], and therefore makes the method more suitable for field applications. In contrast, the magnetic field of a vertical magnetic moment decreases stronger than the cube of the distance over a conductive soil.
- 3) The location is calculated using simple closed-form formulas, enabling real-time localization with only scarce computational power. The closed-form formulas are based on simple expressions for the beacons' signals, which are derived thanks to the particular beacons configuration.
- 4) The values of the beacons magnetic moments are not required for localization calculations, only beacons signals' amplitudes and phases, which are estimated us-

Manuscript received October 17, 2012; revised February 20, 2013; accepted March 9, 2013. Date of publication July 9, 2013; date of current version November 6, 2013. The Associate Editor coordinating the review process was Dr. S. Baglio.

A. Sheinker, B. Ginzburg, and N. Salomonski are with the SOREQ Nuclear Research Center, Yavne 81800, Israel (e-mail: aries@soreq.gov.il; borginz@soreq.gov.il; nizans@soreq.gov.il).

L. Frumkis and B.-Z. Kaplan are with the Department of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Beer-Sheva 84105, Israel. (e-mail: frumkis@ee.bgu.ac.il; kaplan@ee.bgu.ac.il).

Digital Object Identifier 10.1109/TIM.2013.2270919

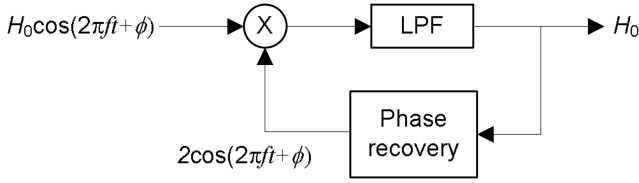


Fig. 1. Lock-in amplifier with phase recovery block is tuned to a specific beacon frequency, in order to estimate beacon signal's amplitude and phase.

ing a lock-in amplifier with a phase recovery block [25].

5) Only a single observation is needed for localization.

For many applications such as indoor or underground mobile robot navigation these features are necessary. Therefore, it is advantageous to use the presently proposed method, since none of the aforementioned techniques in [8]–[19] combines together all these important features.

Numerous computer simulations show good localization accuracy. Tests have shown that the computer simulation can be employed as an effective tool for predicting localization accuracy for various scenarios. Field experiments, conducted with a developed field prototype, validate simulation results. The fair localization accuracy together with simple implementation makes the proposed method attractive for real-time, low-power applications.

II. BACKGROUND

According to Biot-Savart law, a current carrying coil produces a magnetic moment, which is defined as the product of the excitation current I , the number of turns N , and the area A ,

$$\mathbf{M} = A \cdot N \cdot I. \quad (1)$$

The magnetic moment vector, $\mathbf{M}$, is aligned with the coil axis. In this paper we focus on near-field zone, i.e., distances shorter than the electromagnetic wavelength. Therefore, the magnetic flux density, $\mathbf{B}$, is modeled as a dipole field [4], expressed by,

$$\mathbf{B} = (\mu_0/4\pi) \cdot r^{-5} [3(\mathbf{M} \cdot \mathbf{r})\mathbf{r} - M r^2]. \quad (2)$$

The vector from the measurement point to the magnetic moment source is denoted by $\mathbf{r}$, and the permeability of air is denoted by μ_0 . The magnetic field $\mathbf{H}$ is calculated from (2) by,

$$\mathbf{H} = r^{-5} [3(\mathbf{M} \cdot \mathbf{r})\mathbf{r} - M r^2] \quad (3)$$

where the coefficient 4π was omitted from (3) and afterward for simplicity. From (3) we see that the magnitude of the magnetic field is a linear function of the magnetic moment, $\mathbf{M}$. Exciting the coil by an alternating current with frequency f and phase ϕ , results in a time-varying magnetic field,

$$H(t) = H_0 \cos(2\pi f t + \phi). \quad (4)$$

Assuming that the frequency, f , is known, a lock-in amplifier can be used to estimate the magnetic field amplitude, H_0 and phase ϕ . Multiplying the magnetic field, $H(t)$, by a reference

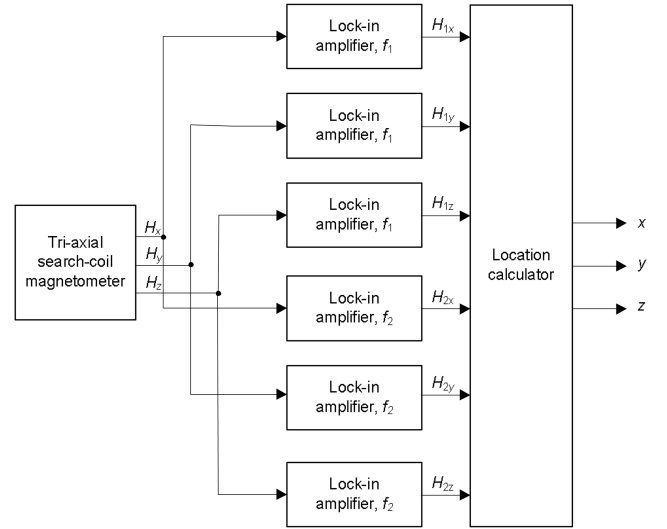


Fig. 2. Basic receiver scheme comprises a tri-axial search-coil magnetometer for low frequencies, six blocks of phase lock-in amplifiers to estimate beacon's signals, and a location calculator. Each triple of lock-in amplifiers is associated with a different beacon, estimating beacons' signals in all three directions. In case the configuration includes three beacons, additional triple of lock-in amplifiers is required, each tuned to the third beacon's frequency.

signal $2\cos(2\pi f t + \phi)$ and applying a low-pass-filter (LPF) with a cutoff frequency lower than $4\pi f$, yields,

$$H_0 = LPF[H_0 \cos(2\pi f t + \phi) \cdot 2 \cos(2\pi f t + \phi)]. \quad (5)$$

From (5) it is clear that the narrower the bandwidth of the LPF, the better selectivity is obtained, enabling the suppression of the magnetic noise and rejection of external interferences of nearby frequencies [5]–[7].

For localization calculation we need accurate measurement of received signal amplitude and its phase relative to beacon's excitation. A lock-in amplifier is employed to achieve this task, as depicted in Fig. 1.

In practice, the excitation frequency can be assigned *a priori* but hardly the phase, which tends to drift. Therefore, we need a phase recovery block to continuously track the phase as a part of a lock-in amplifier. The phase recovery block can be implemented in software by continuously comparing the beacon's signal phase with the reference signal phase, and adjusting the reference signal phase accordingly. Random phase noise influence has been found to be negligible since only the amplitude and sign are required for localization calculations.

When several beacons are used, each beacon can be distinguished by its specific frequency using a properly tuned lock-in amplifier, similarly to the frequency-division multiple access (FDMA) approach used in telecommunication.

Fig. 2 depicts a basic scheme of the receiver, including a tri-axial search-coil magnetometer for low frequencies, and lock-in amplifiers each tuned to a specific beacon frequency. The magnetic fields amplitudes and phases estimated by the lock-in amplifiers for each beacon are then used to localize the receiver.

The dipole model in (3) holds in free space; however, a conductive soil may distort the magnetic field at distances

over hundred meters or more from the beacon [22]–[24]. This can be explained by considering the conductive soil as a conductive half-space with an image dipole. In case of a vertical dipole the magnetic moments have opposite directions, canceling each other, whereas for a horizontal dipole the magnetic moments directions coincide. As a consequence, in this paper, we employ beacons with a horizontal magnetic moment, only.

In case of modest soil conductivity its influence on the beacons' magnetic field in the range of a few hundred meters may be neglected. Therefore the model in (3) holds for the above range of sensor-beacons separation.

On the other hand, in the vicinity of the beacon coil its spatial structure differs from a pure dipole field. Our experimental results (see Section V) show that acceptable localization accuracy is obtained for distances larger than two to threefold the beacon's coil largest dimension.

III. THEORY

A Cartesian coordinate frame is used to represent the beacons' signals, where $\hat{x}$ is a unit vector, aligned with the magnetic North, $\hat{y}$ is a unit vector directed to the West, and $\hat{z}$ is a unit vector directed upward, accordingly.

Consider beacon 1 which is placed at the origin with a magnetic moment M_1 aligned with $\hat{x}$, yielding,

$$\mathbf{M} = M_1 \hat{x}. \quad (6)$$

The location of a sensor at coordinates (x, y, z) can be represented by a vector $\mathbf{r}$,

$$\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}. \quad (7)$$

Substituting (6) and (7) into (3) yields the magnetic field $\mathbf{H}_1$ at sensor location,

$$\mathbf{H}_1 = M_1 r^{-5} [(3x^2 - r^2)\hat{x} + 3xy\hat{y} + 3xz\hat{z}]. \quad (8)$$

From (8) we see that the components of beacon 1 magnetic field measured by the receiver, are given by,

$$H_{1x} = M_1 r^{-5} (3x^2 - r^2) \quad (9)$$

$$H_{1y} = M_1 r^{-5} 3xy \quad (10)$$

$$H_{1z} = M_1 r^{-5} 3xz. \quad (11)$$

Beacon 2 is placed at $(R_x, 0, R_z)$, thus, the vector to the receiver is given by,

$$\mathbf{r}_2 = (x - R_x)\hat{x} + y\hat{y} + (z - R_z)\hat{z}. \quad (12)$$

The magnetic moment of beacon 2, M_2 , is also directed to the $\hat{x}$ axis. Thus, the components of beacon 2 magnetic field measured by the receiver are given by,

$$H_{2x} = M_2 r_2^{-5} [3(x - R_x)^2 - r_2^2] \quad (13)$$

$$H_{2y} = M_2 r_2^{-5} 3(x - R_x)y \quad (14)$$

$$H_{2z} = M_2 r_2^{-5} 3(x - R_x)(z - R_z). \quad (15)$$

Dividing (10) by (11) yields,

$$H_{1y}/H_{1z} = y/z. \quad (16)$$

Now, dividing (14) by (15), yields,

$$H_{2y}/H_{2z} = y/(z - R_z). \quad (17)$$

From (16) and (17) we see that the z coordinate can be calculated using,

$$z = \frac{R_z H_{2y}}{H_{2z} \left(\frac{H_{2y}}{H_{2z}} - \frac{H_{1y}}{H_{1z}} \right)}. \quad (18)$$

Substituting (18) into (16), yields,

$$y = z H_{1y}/H_{1z}. \quad (19)$$

In order to calculate coordinate x we divide (9) by (10), obtaining,

$$\frac{H_{1x}}{H_{1y}} = \frac{3x^2 - r^2}{3xy} = \frac{3x^2 - x^2 - y^2 - z^2}{3xy}. \quad (20)$$

We can rewrite (20) as a quadratic equation,

$$2x^2 - \left(\frac{3yH_{1x}}{H_{1y}} \right) x - y^2 - z^2 = 0. \quad (21)$$

Solving the quadratic equation (21) for x , yields,

$$x = 0.75y \left(\frac{H_{1x}}{H_{1y}} \right) \pm 0.25 \sqrt{9y^2 \left(\frac{H_{1x}}{H_{1y}} \right)^2 + 8(y^2 + z^2)}. \quad (22)$$

In the same fashion coordinate x can be calculated using beacon 2, through dividing (13) by (14). In this case, after manipulation, the result for the x coordinate is expressed by,

$$x = R_x + 0.75y \left(\frac{H_{2x}}{H_{2y}} \right) \pm 0.25 \sqrt{9y^2 \left(\frac{H_{2x}}{H_{2y}} \right)^2 + 8y^2 + 8(z - R_z)^2}. \quad (23)$$

The upper sign in formulas (22) and (23) should be taken for positive values of x , whereas for negative values of x the lower sign should be taken. Hence, taking the proper sign requires some *a priori* knowledge about the receiver location. Adding another beacon eliminates the requirement for *a priori* knowledge and simplifies coordinate x calculation. This configuration is depicted in Fig. 3.

Thus, for beacon 3 located at $(0, R_y, 0)$, the vector to the target is given by,

$$\mathbf{r}_3 = x\hat{x} + (y - R_y)\hat{y} + z\hat{z}. \quad (24)$$

For a magnetic moment M_3 aligned with the $\hat{y}$ direction, the magnetic field components at the receiver are given by,

$$H_{3x} = M_3 r_3^{-5} 3x(y - R_y) \quad (25)$$

$$H_{3y} = M_3 r_3^{-5} [3(y - R_y)^2 - r_3^2] \quad (26)$$

$$H_{3z} = M_3 r_3^{-5} 3(y - R_y)z. \quad (27)$$

Now, coordinate x can be calculated by dividing (25) by (27),

$$x = z H_{3x}/H_{3z}. \quad (28)$$

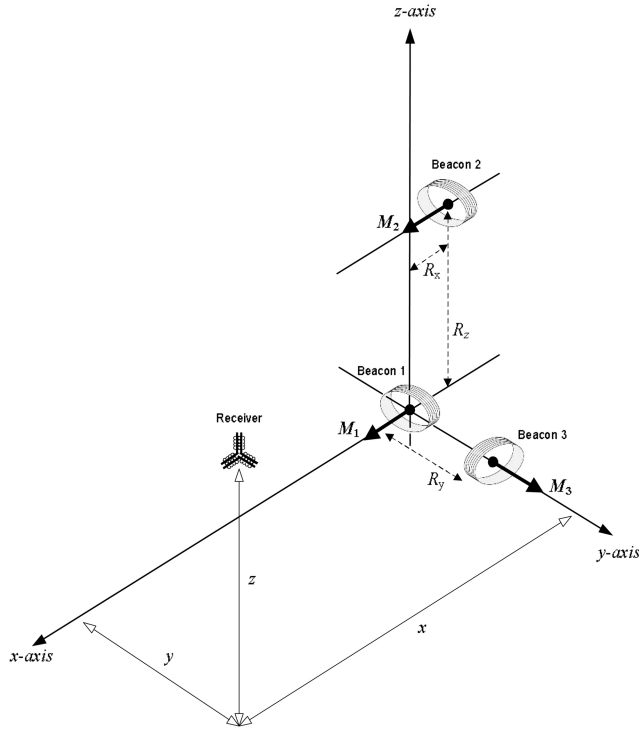


Fig. 3. Proposed method employs beacons 1 and 2 for configurations C1 and C2, and beacons 1, 2, and 3 for configuration C3.

The above analysis enabled the development of the closed-form formulas for localization, which can be simply implemented with only a small computational complexity. Thus, the z and y coordinates have been calculated using (18) and (19), respectively, whereas the x coordinate has been calculated using either (22), (23), or (28). All three aforementioned configurations and the associated formulas are summarized in Table I.

Notice that configurations C1 and C2 rely on different beacons for calculating coordinate x , namely, configuration C1 relies on H_{1x} , whereas C2 relies on H_{2x} .

The localization equations for x , y , and z depend only on the ratio between the measured magnetic fields, which eliminates the need to know *a priori* the value of the magnetic moment.

A. Computer Simulation

Localization accuracy is influenced by various factors such as magnetic noise, magnetic sensor misalignment, and beacons misalignment. Although, an exact error budget analysis is possible, computer simulation seems more suitable, not only because of a number of influencing factors (beacons' magnetic moment, orientation, relative alignment), but due to non-linear dependence of the error as a function of the receiver position. Thus, we have decided to evaluate localization accuracy using a numerical approach, and therefore, developed a dedicated computer simulation. Three simulated magnetic beacons were arranged according to Fig. 3 as summarized by Table II, where the beacons were simulated by point dipole sources.

The simulation uses an area of $10 \text{ m} \times 11 \text{ m}$ size, which is similar in size to our test field. For these ranges the influence of soil conductivity can be neglected, and hence, the model

TABLE I
LOCALIZATION CONFIGURATIONS

Config.	Beacons	Calc. x	Calc. y	Calc. z
C1	1, 2	(22)	(19)	(18)
C2	1, 2	(23)	(19)	(18)
C3	1, 2, 3	(28)	(19)	(18)

TABLE II
SIMULATED BEACONS

Beacon	Magnetic moment	Orient.	Location	Freq.
1	$10 \text{ A}\cdot\text{m}^2$	$\hat{x}$	(0, 0, 0)	630 Hz
2	$10 \text{ A}\cdot\text{m}^2$	$\hat{x}$	(-0.7, 0, 2.2)	535 Hz
3	$10 \text{ A}\cdot\text{m}^2$	$\hat{y}$	(0, -0.7, 0)	770 Hz

in (3) is valid. The area is divided by a 1 m spaced grid, resulting in a total of 110 nodes. The receiver has been placed in air at each node, and the magnetic fields have been calculated using (3) at the points where the receiver has been placed. Since the magnetometer coil length is small (less than 6 cm), the measurement can be considered as being taken at a single point. Afterward, the location was calculated using the proposed methods. The localization error is defined by the Euclidean distance between actual and estimated location. Thus, for a node located at (x, y, z) , and a localization result of $(\tilde{x}, \tilde{y}, \tilde{z})$, the localization error is evaluated by,

$$e(x, y, z) = \sqrt{(x - \tilde{x})^2 + (y - \tilde{y})^2 + (z - \tilde{z})^2}. \quad (29)$$

Obviously, when the beacons and the sensor are oriented properly, and there is no magnetic noise, the calculations show zero localization error. However, in practice, the sensor and the beacons may be misaligned, which together with magnetic noise results in localization errors. Therefore, we use the simulation to test typical scenarios associated with the various factors which govern the localization accuracy, as summarized in Table III.

We have chosen an angular deviation of 1° to test the effect of magnetometer misalignment and beacons misalignment since this accuracy can be easily attained using a magnetic compass and a spirit level. Dedicated geophysical equipment allows the improvement of orientation, even in field conditions. For example in case of scenario 1 the magnetometer has been rotated by 1° around its $\hat{x}$ axis, hence, the orientation of the $\hat{y}$ and $\hat{z}$ axes has been changed by 1° .

We have also tested localization accuracy in noisy environments by adding noise to the simulated beacons' signals. White Gaussian noise with zero mean and variance of 10 pT^2 has been used, which may be considered as a typical noise figure. The procedure of adding noise to the beacons' signals and calculating the localization results has been repeated for 10 000 times.

Scenarios 11, 12, and 13 are used to test the influence of magnetometer axes miss-orthogonality on localization error. Miss-orthogonality values of 1° , 2° , and 5° , have been assumed

TABLE III
LIST OF SIMULATED SCENARIOS

#	Scenario
1	Magnetometer is rotated around $\hat{x}$ by 1°
2	Magnetometer is rotated around $\hat{y}$ by 1°
3	Magnetometer is rotated around $\hat{z}$ by 1°
4	Beacon 1 is rotated around $\hat{y}$ by 1°
5	Beacon 1 is rotated around $\hat{z}$ by 1°
6	Beacon 2 is rotated around $\hat{y}$ by 1°
7	Beacon 2 is rotated around $\hat{z}$ by 1°
8	Beacon 3 is rotated around $\hat{x}$ by 1°
9	Beacon 3 is rotated around $\hat{z}$ by 1°
10	Additive white Gaussian noise
11	Magnetometer axes miss-orthogonality of 1°
12	Magnetometer axes miss-orthogonality of 2°
13	Magnetometer axes miss-orthogonality of 5°

for scenarios 11, 12, and 13, respectively. As expected, we see from Table IV that localization error increases with larger miss-orthogonality. While localization errors are too large for scenario 13, the errors for scenarios 11 and 12 are acceptable in many cases. In practice, miss-orthogonality of 2° or smaller can be obtained by careful assembly of the magnetometer axes. Further improvement can be achieved by magnetometer calibration [29].

The simulation results for receiver height of 1.55 m are summarized in Table IV. We see that the mean localization error for all scenarios (except 13) is less than 0.48 m, whereas for most scenarios it is even much smaller. We also see that the method is especially susceptible to rotation of beacons 1 and 2 around $\hat{y}$, and beacon 3 around $\hat{x}$, making the orientation issue important.

IV. EXPERIMENTS

A field prototype has been developed in order to test the proposed method with real-world signals. The prototype comprises a receiver and three magnetic beacons deployed according to Table II. Each beacon consists of a current carrying coil, a capacitor, and a signal generator [26]. The coil comprises 100 windings of a copper wire on a rectangular wooden frame with a side length of 1 m. The resulting coil inductance is about 16 mH, the resistance is about 5Ω , and the effective area is about 100 m^2 . The capacitor is used to tune the resonance frequency of the RLC circuit determining the specific frequency of the beacon. We have used capacitors values of 4, 5.5, and $2.7\mu\text{F}$ to obtain frequencies of 630, 770, and 535 Hz, according to Table II. The signal generator is used to drive the coil with a current of 0.1 A at the resonance frequency, producing a magnetic moment of about $10\text{ A}\cdot\text{m}^2$.

The receiver is based on a tri-axial search-coil magnetometer for low frequencies, with frequency bandwidth of 0.5–3 kHz. Within mentioned frequency bandwidth an intrinsic magnetic noise is less than $0.6\text{ pT}/\sqrt{\text{Hz}}$. In particular, at

TABLE IV
SIMULATION RESULTS FOR THE LISTED SCENARIOS

Scenario	Localization error, m					
	Config. C1		Config. C2		Config. C3	
	Mean	Max	mean	Max	Mean	Max
1	0.12	0.19	0.13	0.20	0.39	1.71
2	0.11	0.57	0.12	0.58	0.39	0.91
3	0.11	0.33	0.12	0.34	0.10	0.33
4	0.44	1.65	0.46	1.67	0.36	1.62
5	0.09	0.26	0.11	0.25	0.08	0.24
6	0.46	1.64	0.48	1.66	0.46	1.64
7	0.04	0.10	0.10	0.26	0.04	0.10
8	-	-	-	-	0.42	1.78
9	-	-	-	-	0.09	0.25
10	0.14	0.64	0.14	0.66	0.17	0.79
11	0.13	0.21	0.13	0.24	0.29	0.64
12	0.27	0.43	0.26	0.48	0.56	1.27
13	0.68	1.04	0.67	1.16	1.29	3.10

535 Hz, the lowest frequency used, the spectral noise density is found to be about $0.33\text{ pT}/\sqrt{\text{Hz}}$. The sensing coils with ferromagnetic (permalloy) cores are arranged inside a cube with a side length of 6 cm. It is guaranteed by the manufacturer that the axes miss-orthogonality is smaller than 5° , and in fact it turns out to be even smaller than 2° . These characteristics have been provided by the manufacturer [27] and approved by our measurements. Magnetometers with similar characteristics are also described in literature [28].

The sensor outputs are sampled at a rate of 25 000 samples/s using the 24 bit data acquisition unit, NI USB-4431. The readings are then processed by a PC which is connected to the NI USB-4431 via a USB connection. The PC runs a dedicated LABVIEW application which includes the lock-in amplifiers and the localization calculations for the various configurations in Table I. The lock-in amplifiers include a third order LPF with a cutoff frequency of 10 Hz. A time span of 1 s is used for the lock-in amplifier window to estimate the beacons' signals. The setup is depicted in Fig. 4.

The beacons have been deployed in the test field according to Table II. The beacons have been placed with accuracy of about 0.15 m, and their orientation has been set using a simple compass. The receiver was installed on a wheeled cart, enabling to vary the receiver's height and move the receiver around the test field. The test field, with an area size of $10\text{ m} \times 11\text{ m}$, has been divided by a 1 m spaced grid, similarly to the computer simulation. During the test, the receiver has been placed at each node of the grid with a placement error of about 0.15 m, and then, the location was calculated using the proposed methods. The measurements were repeated for two different sensor heights, 0.55 and 1.55 m in the air above beacons centers. The distances are measured between physical centers of the magnetometer and the beacons using a laser range finder and a ruler. In practice, the offset between the physical and the magnetic center is negligible. Fig. 5 depicts the error map obtained for the proposed localization method in configuration C1, where outliers along the axes and nearby the beacons at the origin were removed. Localization errors near the beacons are expected because the magnetic



Fig. 4. Experimental setup: the magnetic beacons (1) and (2); the wheeled cart (3); the receiver (4) including a LEMI-131 search coil magnetometer, a data acquisition unit NI USB-4431, and a battery pack; a laptop PC (5) running the dedicated software application implementing the proposed method in real-time.

field at distances smaller than two to threefold the beacon's largest dimension differs substantially from a pure dipole field structure.

The experimental results are summarized in Table V, where in rows 1 and 2 the localization errors are calculated using (29). To further reduce localization errors, we applied bias removal. Such a bias can be caused by beacons misplacement and misalignment. These factors, however, are constant over the entire field test, in contrast to magnetic noise and sensor misalignment which may vary. The bias $\mathbf{d}$ is defined as a vector which minimizes the mean localization error,

$$\bar{\mathbf{e}} = \frac{1}{N} \sum_{k=1}^N \|\mathbf{r}_k - \tilde{\mathbf{r}}_k - \mathbf{d}\|. \quad (30)$$

Here, $\mathbf{r}_k$ is the actual location of node k , $\tilde{\mathbf{r}}_k$ is the localization result, and N is the total number of nodes. In fact, the bias can be estimated in advance for a given beacons setup as a part of system calibration procedure (in this case, it is sufficient to use only a limited number of nodes for bias calculation). Thus, experiment result show that the localization error is smaller after bias removal as summarized in Table V, rows 3 and 4.

TABLE V
EXPERIMENTAL RESULTS

Rec. height (m)	Localization error, m						
	Config. C1		Config. C2		Config. C3		
	mean	max	mean	max	mean	max	
0.55	0.25	0.77	0.50	0.88	0.83	1.77	
1.55	0.23	0.60	0.42	0.94	0.75	1.68	
0.55	0.25	0.63	0.26	0.62	0.80	1.83	after bias removal
1.55	0.14	0.47	0.21	0.53	0.50	1.27	

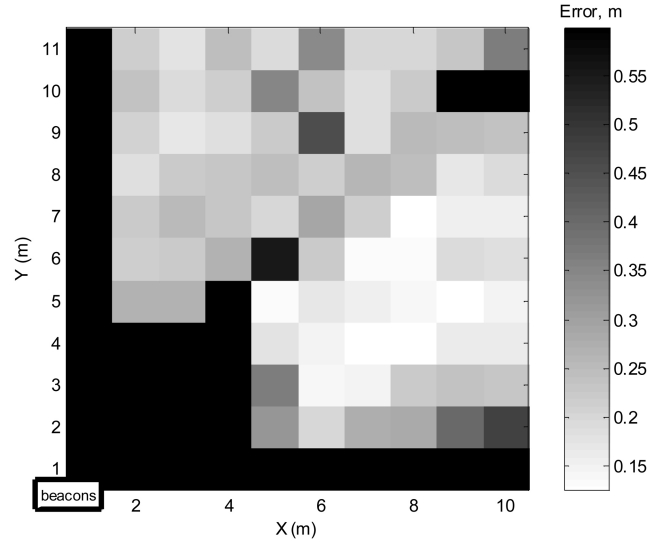


Fig. 5. Localization error map obtained from the experiments, after removing the bias and outliers along the axes and nearby the beacons at the origin.

From Table V, we see that for configuration C1, the mean localization error, obtained from experiment results without bias removal, varies between 0.23 and 0.25 m, and the maximal localization error varies between 0.60 and 0.77 m. These results are in the range of values expected from the simulations in Table IV. Obviously, after bias removal, the mean and maximal localization errors are smaller, as expected. Notice, that configurations C1 and C2 demonstrate smaller localization errors than configuration C3. This was already presented in Table IV using the simulation. Configuration C3 employs an extra beacon, which increases the errors due to possible beacon misplacement and misalignment. One should also notice that configuration C1 demonstrates smaller localization errors than configuration C2, which is barely noticeable in Table IV. The reason is that location calculations in configuration C1 are done using (22), whereas location calculations in configuration C2 are done using (23), which requires extra parameters R_x and R_z . In practice, the values of R_x and R_z are measured with an accuracy of about 0.15 m, which introduces errors in location calculations for configuration C2.

The obtained results can be compared with the results obtained in [21] for the 2-D case, where a similar localization approach has been proposed. The methods in the 2-D and 3-D cases differ by beacons configurations, where for the 3-D case estimating receiver elevation requires different beacons elevations. In addition, it is sufficient to use a dual axis magnetometer for the 2-D case, whereas for the 3-D case a

tri-axis magnetometer is required. A mean localization error of 0.13 m and a maximal localization error 0.34 m have been obtained for the most accurate configuration in the 2-D case. Comparison with configuration C1 shows, that although the results are close, the localization accuracy in the 2-D is higher. This can be explained using the fact that in most cases estimation of three variables is more erroneous than the estimation of only a pair of variables. Thus, in the horizontal plain, the mean localization error of C1 is 0.14 m and the maximal error is 0.38 m, which are very close to the results obtained for the 2-D case.

Comparing the experimental results with the computer simulation shows that the latter can be exploited as a useful tool to predict the performance of localization for different SNR values and various scenarios. The obtained small localization errors are acceptable in variety of cases, making the proposed method attractive for many applications.

V. FURTHER INVESTIGATION

The localization results obtained for the proposed method may be improved by more accurate orientation of the beacons and calibration of the search coil magnetometer [29]. The longer the localization range the more pronounced is the influence of these accuracies on localization results.

Further improvement in localization accuracy may be obtained by using magnetic, inertial, or other types of direction finder, which enables to calculate accurately the projections of the magnetometer readings onto the defined coordinate frame [30], [31].

Future investigation will include a method for magnetic localization without the need to orient the three-axial magnetometer, which is a great benefit when on-the-move localization is required. This method may rely on calculating the magnetic total-field from the three-axial readings and searching for the fittest solution using one of the well-known global search methods [32]. Although consuming a lot of computational power, preliminary tests show that this numerical approach is very effective.

VI. CONCLUSION

In this paper we proposed a method for localizing a magnetic receiver using beacons of low frequency magnetic field. The method was tested using numerous computer simulations showing accurate localization results in various scenarios. A field prototype of the system was developed and tested in field conditions, validating simulation results. Configurations C1 and C2 showed similar results, where configuration C1 is a little more accurate—localization error less than 0.77 m and a mean value of 0.25 m. Although configuration C3 was less accurate than configurations C1 and C2 it required less calculations and did not require *a priori* knowledge of the quadrant where the receiver is located. Nevertheless, all configurations presented fair localization accuracy, which together with the simple implementation make the proposed method attractive for many real-time, low power applications requiring high field penetration ability, such as indoor robot navigation, underground cavity mapping, and many more.

ACKNOWLEDGMENT

The authors would like to thank Y. Elek, E. Shviro, A. Noyman, and A. Yaniv for their important technical assistance.

REFERENCES

- [1] R. Barr, D. L. Jones, and C. J. Rodger, "ELF and VLF radio waves," *J. Atmos. Sol. Terr. Phys.*, vol. 62, nos. 17–18, pp. 1689–1718, Nov. 2000.
- [2] B. Gozick, K. P. Subbu, R. Dantu, and T. Maeshiro, "Magnetic maps for indoor navigation," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 12, pp. 3883–3891, Dec. 2011.
- [3] A. D. Ash, "Noise and noise reduction techniques for airborne magnetic measurements at sea," in *Proc. Int. Conf. Marine Electromagn. (MARELEC)*, 1997, p. 4.
- [4] R. J. Blakely, *Potential Theory in Gravity and Magnetic Applications*. Cambridge, U.K.: Cambridge Univ. Press, 2001.
- [5] M. A. Nixon, B. C. McCallum, and N. B. Price, "The effects of metals and interfering fields on electromagnetic trackers," *Presence*, vol. 7, no. 2, pp. 204–218, Apr. 1998.
- [6] V. F. Labson, A. Becker, H. F. Morrison, and U. Conti, "Geophysical exploration with audiofrequency natural magnetic fields," *Geophysics*, vol. 50, no. 4, pp. 656–664, Apr. 1985.
- [7] F. H. Raab, "Noise model for low-frequency through-the-earth communication," *Radio Sci.*, vol. 45, no. 6, pp. 1–7, Dec. 2010.
- [8] F. H. Raab, E. B. Blood, T. O. Steiner, and H. R. Jones, "Magnetic position and orientation tracking system," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 15, no. 5, pp. 709–718, Sep. 1979.
- [9] M. C. Domingo, "Magnetic induction for underwater wireless communication networks," *IEEE Trans. Antennas Propag.*, vol. 60, no. 6, pp. 2929–2939, Jun. 2012.
- [10] A. Tran, M. Bolic, and M. C. E. Yagoub, "Magnetic-field coupling characteristics of ferrite-coil antennas for low-frequency RFID applications," *Int. J. Comput. Sci. Issues*, vol. 7, no. 4, pp. 7–11, Jul. 2010.
- [11] C. P. Davis, W. C. Chew, W. W. Tucker, and P. R. Atkins, "A null-field method for estimating underground position," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3731–3738, Nov. 2008.
- [12] J. Sodage, Y. Vichabian, A. Vandiver, P. Reppert, D. Coles, and F. D. Morgan, "Electromagnetic cave-to-surface mapping system," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 4, pp. 754–763, Apr. 2004.
- [13] A. Markham, N. Trigoni, D. W. Macdonald, and S. A. Ellwood, "Underground localization in 3-D using magneto-inductive tracking," *IEEE Sens. J.*, vol. 12, no. 6, pp. 1809–1816, Jun. 2012.
- [14] S. Hashi, M. Toyoda, S. Yabukami, K. Ishiyama, Y. Okazaki, and K. I. Arai, "Wireless magnetic motion capture system—Compensatory tracking of positional error caused by mutual inductance," *IEEE Trans. Magn.*, vol. 43, no. 6, pp. 2364–2366, Jun. 2007.
- [15] A. Plotkin and E. Paperno, "3-D magnetic tracking of a single subminiature coil with a large 2-D array of uniaxial transmitters," *IEEE Trans. Magn.*, vol. 39, no. 5, pp. 3295–3297, Sep. 2003.
- [16] G. Placidi, D. Franchi, A. Maurizi, and A. Sotgiu, "Review on patents about magnetic localization systems for in vivo catheterizations," *Recent Patents Biomed. Eng.*, vol. 2, no. 1, pp. 58–64, Jan. 2009.
- [17] N. H. Nessler, "Electromagnetic location system for trapped miners," in *Proc. SPIE Conf. Subsurface Sens. Appl.*, vol. 3752, Jul. 1999, pp. 252–263.
- [18] H. Qi, J. Ye, X. Zhang, and H. Chen, "Wireless tracking and locating system for in-pipe robot," *Sens. Actuators A, Phys.*, vol. 159, no. 1, pp. 117–125, Apr. 2010.
- [19] N. Ayuso, J. A. Chuchi, F. Lera, and J. L. Villarreal, "Accurately locating a vertical magnetic dipole buried in a conducting earth," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3676–3685, Oct. 2010.
- [20] S. K. Harriman, E. W. Evans, W. Paschal, and U. S. Inan, "Magnetic sensor design for femtoTesla low-frequency signals," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 1, pp. 396–402, Jan. 2010.
- [21] A. Sheinker, B. Ginzburg, N. Salomonski, L. Frumkis, and B. Z. Kaplan, "Localization in 2-D using magnetic beacons of low frequency magnetic fields," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, part 3, pp. 1020–1030, Apr. 2013.
- [22] M. L. Burrows, *ELF Communications Antennas*. London, U.K.: P. Peregrinus Ltd. (On behalf of the Institution of Electrical Engineers), 1978.
- [23] D. H. S. Cheng, "The reflected impedance of a circular coil in the proximity of a semi-infinite medium," *IEEE Trans. Instrum. Meas.*, vol. 14, no. 3, pp. 107–116, Sep. 1965.

- [24] J. R. Wait, "The magnetic dipole over the horizontally stratified earth," *Can. J. Phys.*, vol. 29, no. 6, pp. 577–592, Nov. 1951.
- [25] H. C. So, Y. T. Chan, Q. Ma, and P. C. Ching, "Comparison of various periodograms for sinusoid detection and frequency estimation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 35, no. 3, pp. 945–952, Jul. 1999.
- [26] B. Cheng, A. N. Abdalla, X. Huang, T. H. Lan, and J. Lin, "Design of CHWHG type low frequency magnetic fields generator," *J. Comput. Sci.*, vol. 3, no. 2, pp. 81–83, 2007.
- [27] Lviv Centre of Institute of Space Research [Online]. Available: <http://www.isr.lviv.ua>
- [28] C. Coillot, J. Moutoussamy, R. Lebourgeois, S. Ruocco, and G. Chanteur, "Principle and performance of a dual-band search coil magnetometer: A new instrument to investigate fluctuating magnetic fields in space," *IEEE Sens. J.*, vol. 10, no. 2, pp. 255–260, Feb. 2010.
- [29] E. Pulz, "A calibration facility for search coil magnetometers," *Meas. Sci. Technol.*, vol. 13, no. 5, pp. N49–N51, May 2002.
- [30] D. Roetenberg, H. Luinge, and P. Veltink, "Inertial and magnetic sensing of human movement near ferromagnetic materials," in *Proc. 2nd IEEE ACM Int. Sympos. Mixed Augment. Reality*, 2003, pp. 268–269.
- [31] S. Bonnet, C. Bassompierre, C. Godin, S. Lesecq, and A. Barraud, "Calibration methods for inertial and magnetic sensors," *Sens. Actuators A, Phys.*, vol. 156, no. 2, pp. 302–311, Dec. 2009.
- [32] S. Y. Yang, S. L. Ho, G. Z. Ni, J. M. Machado, and K. F. Wong, "A new implementation of population based incremental learning method for optimizations in electromagnetics," *IEEE Trans. Magn.*, vol. 43, no. 4, pp. 1601–1604, Apr. 2007.



Arie Sheinker (M'06) was born in Czernowitz, USSR (now Ukraine), in 1971. He received the B.Sc., M.Sc., and Ph.D. degrees from the Department of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Israel, in 1992, 2003, 2009, respectively.

He served as a Communications Systems Engineering Officer in the Israeli air force from 1993 to 1999. Since 2004, he has been with the SOREQ Nuclear Research Center, Israel. His main research is in the field of magnetometry, including methods for

magnetic anomaly detection, methods for localization and magnetic moment estimation, magnetic signal processing, magnetic sensing applications, and wireless networks of magnetic sensors. Currently, his research is focused on low frequency magnetic fields and their applications in tracking and communication.



Boris Ginzburg was born in St. Petersburg, Russia, in 1951. He received the M.Sc. degree in radio-physics and electronics from the Technical University of St. Petersburg, St. Petersburg, in 1974, and the Ph.D. degree in physics and mathematics from the Phys.-Techn. Ioffe Institute, St. Petersburg, in 1986.

He was a Research Scientist with the Geophysical Research Institute, St. Petersburg, Russia, from 1974 to 1996. During this period, he was engaged in research and development of optical pumping

magnetometers for precise measurements of the Earth's magnetic field. His experimental investigation of optical pumping phenomena in alkalis and helium resulted in design of new Rb–He and K–He magnetometers with very high metrological features. He headed a research group concerned with elaboration of technological processes for precise magnetic sensors production. Currently he is the Head of a research group at the Nuclear Research Center SOREQ, Israel. His current research interests include precise measurements of the Earth's magnetic field and various magnetic search and detection applications.



Nizan Salomonski (M'09) was born in Haifa, Israel, in 1963. He received the B.Sc., M.Sc., and Ph.D. degrees in mechanical engineering from the Technion - Israel Institute of Technology, Haifa, Israel, in 1991, 1994, and 1999, respectively.

From 1991 to 1994, he was an Assistant Scientist with the Robotics Laboratory, Technion. During this period, he was engaged in research and development of flexible inflatable articulated robots and control algorithms related to the tool trajectory. From 1994 to 1999, he was a Scientist with the Center for Manufacturing Systems and Robotics (CMSR), Technion. During this period, he was engaged in research and development of nonparametric algorithms for adaptive disassembly processes and planning mechanisms. Since 1999, he has been a Researcher at the SOREQ Nuclear Research Center, Israel. Currently, he is the Head of the Research and Development Integrated System Group, SOREQ Nuclear Research Center. His current research interests include nonparametric algorithms for prediction, detection, and localization, and their applications for various magnetic anomaly detection systems.

Dr. Salomonski has been a member of the American Society for Mechanical Engineering since 1998.



Lev Frumkis was born in 1938 in Barnaul, Russia. He received the M.Sc. degree in radio-physics and electronics in 1961, the D.Phil. degree in radio-physics in 1967, and the D.Sc. degree in radio-physics in 1989, all from Tomsk State University, Russia.

He served as a Scientist at the Siberian Physics and Technology Institute, Tomsk, Russia, where he worked on various electromagnetic problems from 1961 to 1990. He immigrated to Israel in 1991.

He served as an Engineer with the Israel Aircraft

Industry from 1991 to 1993. He has been with the Department of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Beer-Sheva, Israel, since 1994, where he is currently a Research fellow financed by a Ministry of Absorption Grant. His current research interests include the fields of magnetic shielding and magnetometry.



Ben-Zion Kaplan (M'76–SM'82–LSM'07) was born in Tel-Aviv, Israel, in 1936. He received the B.Sc. (*cum laude*) and M.Sc. degrees from the Technion-Israel Institute of Technology, Haifa, Israel, in 1958 and 1964, respectively, and the D.Phil. degree in electrical engineering from the University of Sussex, Falmer, Brighton, U.K., in 1971.

From 1961 to 1968, he was a Research Engineer at the Electronics Department (presently the Department of Physics of Complex Systems), Weizmann Institute of Science, Rehovot, Israel. From 1968 to

1971, he was with the School of Applied Sciences, Inter University Institute of Engineering Control, University of Sussex. Since 1972, he has been with the Department of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Beer-Sheva, Israel, where he has been a Professor since 1985 and a Professor Emeritus since 2006. He established the Laboratory for Magnetic and Electronic Systems. He was the incumbent of the Chinita and Conrad Abrahams–Curiel Chair in Electronic Instrumentation from 1988 to 2006. In 1992, he was on sabbatical leave in the Department of Physics, University of Otago, Dunedin, New Zealand. He has published more than 140 articles in refereed scientific journals. His current research interests include magnetic and electronic instrumentation, electromechanical devices, including magnetic levitators and synchronous machines, nonlinear phenomena in electronic networks and magnetic devices, nonlinear and chaotic oscillations, coupled oscillator systems, multiphase oscillators, magnetometry and its relationship to ELF phenomena, magnetic and electric fields sensors for DC and ULF.

Dr. Kaplan is a fellow of the Electromagnetic Academy, a member of the Israeli Committee of URSI, and a member of its Metrology Subcommittee. He obtained a prize in the field of applied electronics donated by the Polish-Jewish Ex-Servicemen's Association, London, in 1993. The prize was due to his achievements in nonlinear electronics and in magnetics.